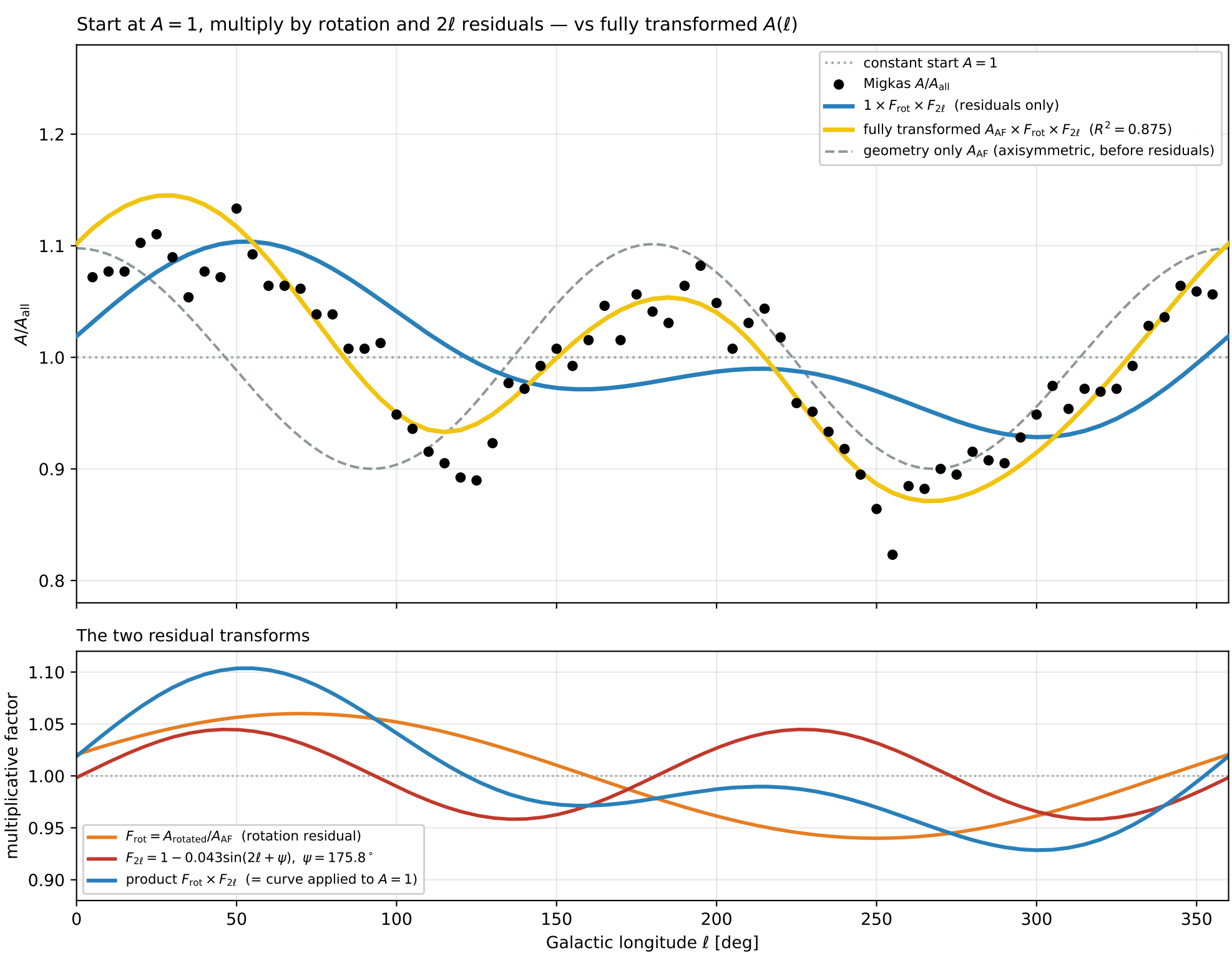
ST+CoE: constant $A = 1$ × (rotation residual) × (2ℓ residual)
Start at $A = 1$, multiply by rotation and 2ℓ residuals — vs fully transformed $A(\ell)$
constant start $A = 1$
Migkas $A/A_{\rm all}$
$1 \times F_{\rm rot} \times F_{2\ell}$ (residuals only)
fully transformed $A_{\rm AF} \times F_{\rm rot} \times F_{2\ell}$ ($R^2 = 0.875$)
geometry only $A_{\rm AF}$ (axisymmetric, before residuals)
$A/A_{\rm all}$
1.2
1.1
1.0
0.9
0.8
The two residual transforms
$F_{\rm rot} = A_{\rm rotated}/A_{\rm AF}$ (rotation residual)
$F_{2\ell} = 1 - 0.043\sin(2\ell + \psi)$, $\psi = 175.8^\circ$
product $F_{\rm rot} \times F_{2\ell}$ (= curve applied to $A = 1$)
multiplicative factor
1.10
1.05
1.00
0.95
0.90
0
50
100
150
200
250
300
350
Galactic longitude ℓ [deg]